

Production & Costs

Basic Cost Concepts

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- ▶ **Opportunity costs:** The cost of a good measured by the alternative uses that are foregone by producing the good.
- †
- ▶ **Accounting cost:** The actual cost paid for inputs, or alternatively the costs for which receipts are available.
- ▶ **Economic cost:** The amount required to keep an input in its present use or the amount that input would be worth in its next best alternative use.
- ▶ Includes opportunity costs. Example - inputs not paid for.
 - ▶ Does **NOT** include sunk costs.

Example

Difference between accounting and economic costs

Suppose you own a beach house that you typically rent out for \$100 a day. You are deciding whether to go on a vacation to the beach house or not for the weekend (2 days). Suppose if you were to go for the vacation, then the expenses would be: travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400). What are accounting and economic costs of the vacation?

$$\text{Acc. Cost} = 800 + 400 = 1200.$$

$$\begin{aligned}\text{Ec. Cost} &= \text{Ac. Cost} + \text{Opp. Cost} \\ &= 1200 + (2 \times 100) \\ &= 1400.\end{aligned}$$


Example

Difference between accounting and economic costs

Suppose you own a beach house that you typically rent out for \$100 a day. You are deciding whether to go on a vacation to the beach house or not for the weekend (2 days). Suppose if you were to go for the vacation, then the expenses would be: travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400). What are accounting and economic costs of the vacation?

Accountant's version of costs: They would infer that the cost of such a vacation would be a total of \$1200, which consists of travel expenses (\$800) and other miscellaneous costs related to food and other supplies (\$400).

Economists' version of costs: They would estimate that the total cost as \$1400. This would include the *opportunity cost* of not being able to rent out the beach house for two days (friday-sunday), which is equal to the potential money that could have been earned if it had been rented out for that period (= \$200).

Labor Vs. Capital Costs

Costs of different inputs in production

2 inputs.

$$Y = f(X_1, X_2)$$

labor. $\swarrow$ L , $\searrow$ K $\swarrow$ Capital
machinery. $\swarrow$ (money)

Labor Vs. Capital Costs

Costs of different inputs in production

Labor Costs

To the accountant, firms spending on wages and salaries is a current expense.

Economists regard wage payments as an *explicit* cost, wherein labor services are purchased at a cost (wage rate).

Both economists & accountants view this **similarly**.

Labor Vs. Capital Costs

Costs of different inputs in production

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Capital Costs

Accountants use historical price of capital and apply depreciation rules.

Economists calculate the *implicit* cost of capital by focussing on what others might pay to use it (rental rate).

Economists & accountants view this **differently**.

Handwritten notes: $\$$ 20,000, 2000, 10 years.

Total Costs

$$\begin{aligned} TC &= \text{cost of } L + \text{cost of } K \\ &= \underbrace{w \cdot L}_{\text{wage}} + \underbrace{v \cdot K}_{\text{rental rate}} \end{aligned}$$

→ $\underbrace{wL + vK = TC}_{\text{straight line}}$ numbers/constants.

Total Costs

Assumptions:

- ▶ There are two inputs: Labor (L) and Capital (K).
- ▶ The input prices are given and constant.

Given these assumptions, total cost (TC) can be expressed as:

$$TC = \underbrace{wL}_{\text{Labor Costs}} + \underbrace{vK}_{\text{Cost of Capital}}$$


Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

$$\underbrace{TC}_{\text{fix}} = \underbrace{wL}_{\text{numbers}} + \underbrace{vK}_{\text{numbers}}$$

Straight
line

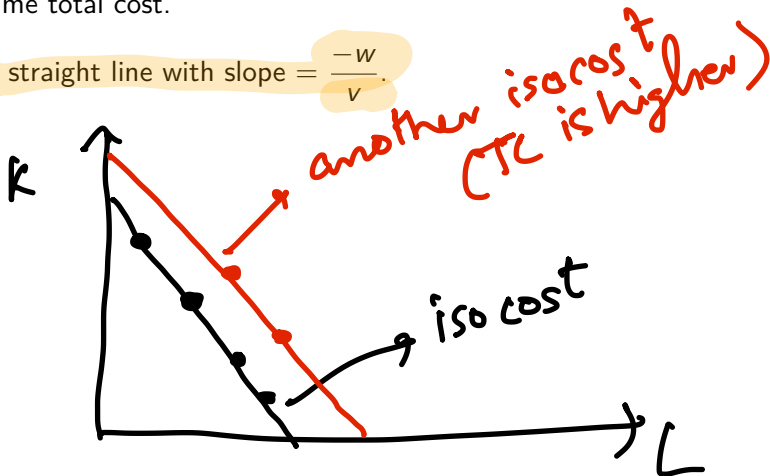
$$\underbrace{\text{constant} = wL + vK}_{\text{slope} = -\frac{w}{v}}$$

Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

It is a straight line with slope = $-\frac{w}{v}$.



Isocost

$$TC = wL + vK$$

Isocost curve represents all combinations of L and K that lead to the same total cost.

It is a straight line with slope $= \frac{-w}{v}$.

This is akin to the budget constraint. (Think why!)

Economic Profits $\rightarrow$ max. Profits.

$$\text{Profits}(\pi) = \text{Total Revenue} - \text{Total Costs}$$

$$= P \cdot q - TC$$

$$= P \cdot q - [wL + rK]$$

$$= Pq - wL - rK$$

Economic Profits

$$\text{Profits}(\pi) = \text{Total Revenue} - \text{Total Costs}$$

If the firm's production function is given by $q = f(K, L)$, then the total revenues of the firm can be expressed as $P.f(K, L)$, where P is the per-unit price of the good produced (output). Therefore, profits (π) can be calculated using:

$$\text{Profits}(\pi) = P.f(K, L) - wL - vK$$

Profit maximization

Given a particular output level

Known: w, v, q_1

Unknown: L, K

$$\text{Profits}(\pi) = P \cdot f(K, L) - wL - vK$$

We start with the assumption that the firm assumes a certain output level (say q_1). Then,

Profits =
(maximize)

$P \cdot q_1$
Constant

$- wL - vK$
minimized

Profit maximization

Given a particular output level

$$Profits(\pi) = P \cdot f(K, L) - wL - vK$$

We start with the assumption that the firm assumes a certain output level (say q_1). Then,

$$Profits(\pi) = P \cdot q_1 - wL - vK$$

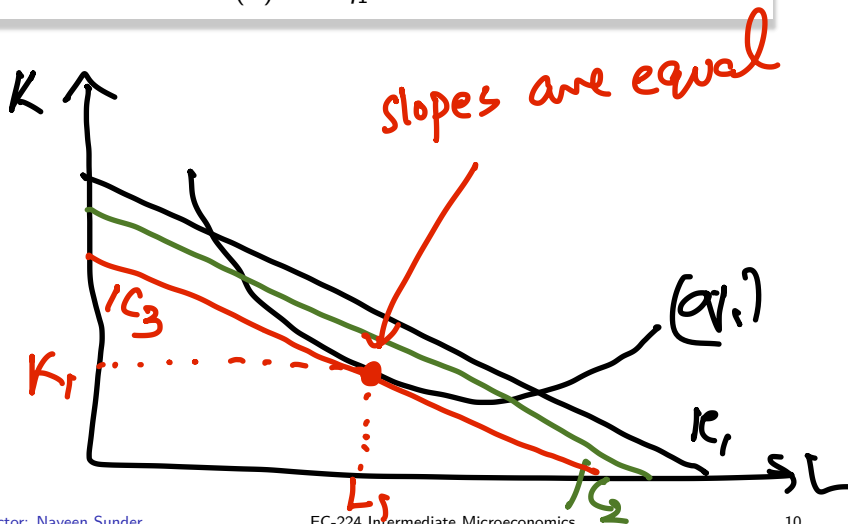
Therefore, given an output level of q_1 , the firm can maximize profits by minimizing costs. The profit maximization problem can be seen as a cost minimization one.

Cost Minimization

Graphical approach

isocost : min. cost

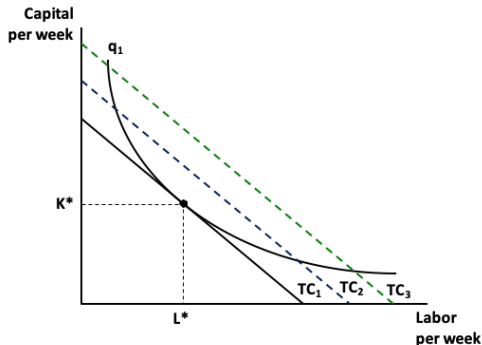
$$\text{Profits}(\pi) = P \cdot q_1 - wL - vK$$



Cost Minimization

Graphical approach

$$Profits(\pi) = P \cdot q_1 - wL - vK$$



The cost minimization problem is graphically equivalent to finding the (K, L) combination on the isoquant for quantity q_1 where the cost of production is minimized.

Cost Minimization

Optimality condition

$$\underbrace{\text{Rate of technical substitution (RTS) of L and K}}_{\text{-Slope of Isoquant}} = \underbrace{\frac{-w}{v}}_{\text{Slope of isocost}}$$

where w = wage rate (per unit cost labor cost),
and v = per unit cost of capital

Cost Minimization

Optimality condition


$$\frac{f_{MP_L}}{MP_K} = \frac{fw}{v}$$

Rate of technical substitution (RTS) of L and K = $\frac{-w}{v}$
-Slope of Isoquant Slope of isocost

where w = wage rate (per unit cost labor cost),
and v = per unit cost of capital

Equilibrium: $\frac{MP_L}{MP_K} = \frac{w}{v}$

Cost Minimization

"Bang for the buck" interpretation

Benefit of 1
move L

$$\frac{MP_L}{w} = \frac{MP_K}{v}$$

Benefit of one
move K

cost of one
move L

cost of one
move K

Cost Minimization

"Bang for the buck" interpretation

$$\frac{MP_L}{w} = \frac{MP_K}{v}$$

This condition implies that the producer should use K and L in such a way that they receive the same "bang for the buck".

Explanation: This approach can alternatively be thought of as equating the "per-dollar" benefit of additional labor with that of additional capital. The gain from using one more unit of labor is the additional output produced by it, which is equal to MP_L . The costs of using one more unit of labor are the wages paid, which is equal to w . Therefore, the "per-dollar" value of this additional unit of labor is $\frac{MP_L}{w}$. The same is the case for capital.

Total Cost function

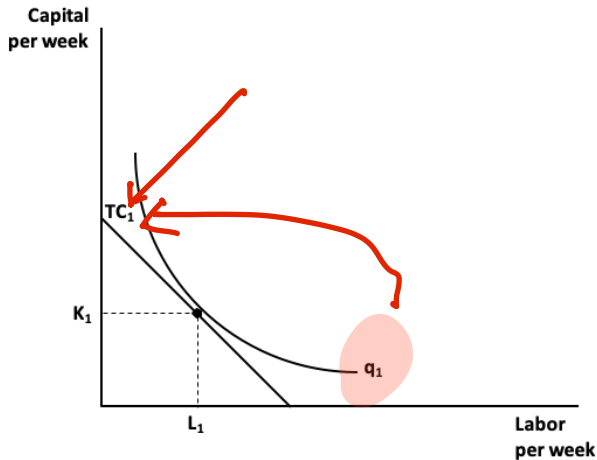
Suppose we start with a production level of q_1

$TC = wL + vk$

$q_1 \xrightarrow[\text{cost}]{\text{min.}} TC_1$

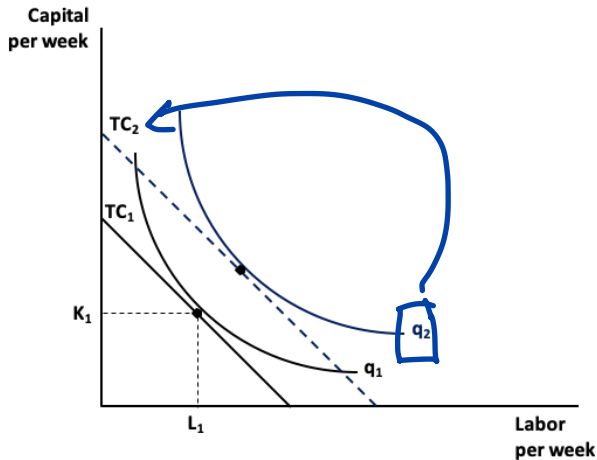
Total Cost function

Suppose we start with a production level of q_1



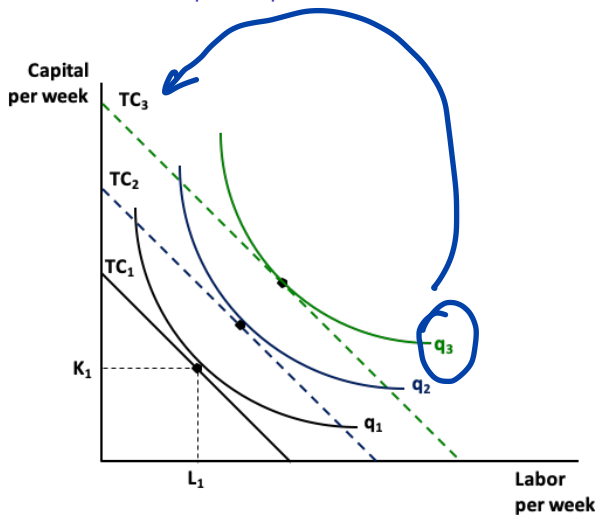
Now we want to shift to output level q_2

Then do cost minimization for output = q_2



Now we want to increase output to q_3

Then do cost minimization for output = q_3

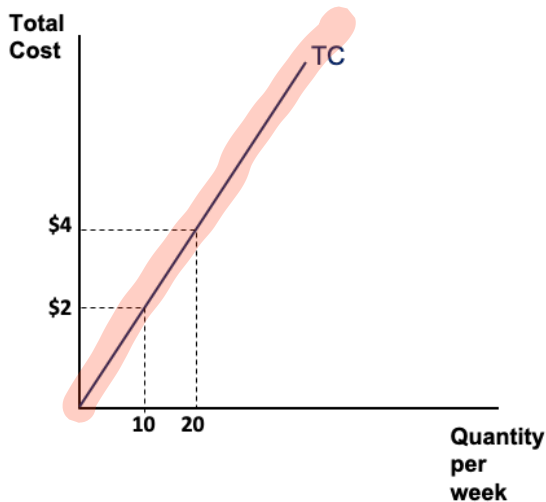


Firm's total cost (TC) curve

In the current case, the output and costs increase at the same rate
- thus this is a situation with constant returns to scale.

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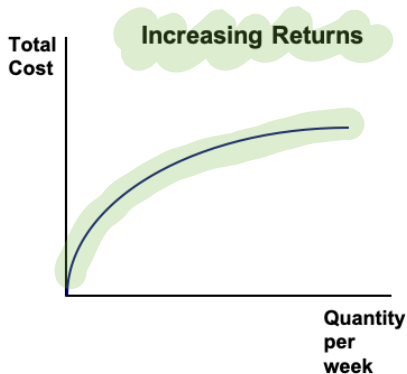


Firm's total cost (TC) curve

Increasing/Decreasing Returns

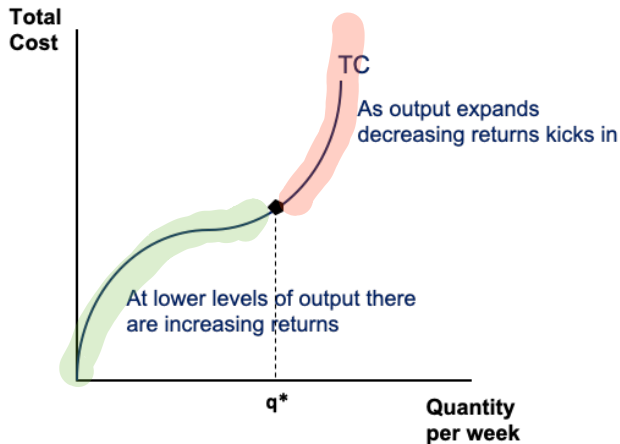
Firm's total cost (TC) curve

Increasing/Decreasing Returns



Firm's total cost (TC) curve

Firm with efficient scale – combining most scenarios.



Average and Marginal costs

Average and Marginal costs

$$\text{Average cost (AC)} = \frac{\text{Total Cost}}{\text{Quantity}} = \frac{TC}{q}$$

This is the per-unit cost of production.

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$$\text{Average cost (AC)} = \frac{\text{Total Cost}}{\text{Quantity}} = \frac{TC}{q}$$

This is the per-unit cost of production.

$$\text{Marginal cost (MC)} = \frac{\text{Change in Total Costs}}{\text{Change in Quantity}} = \frac{\Delta TC}{\Delta q}$$

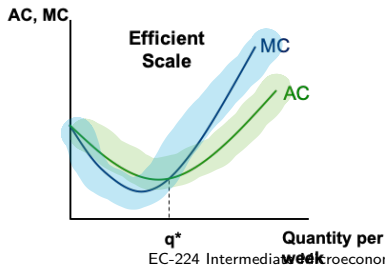
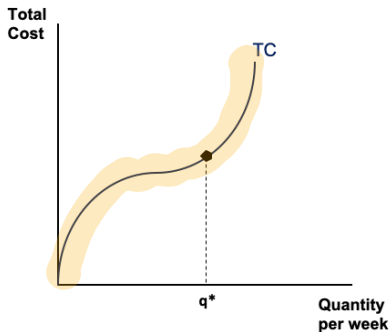
Marginal cost is the additional cost of producing one more unit of output. It is represented by the slope of the total cost curve.

Firms with efficient scale

Shape of AC and MC curves

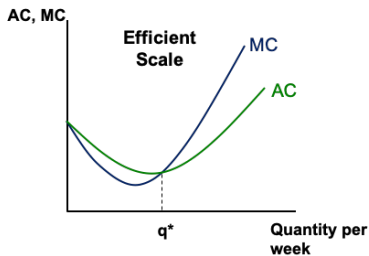
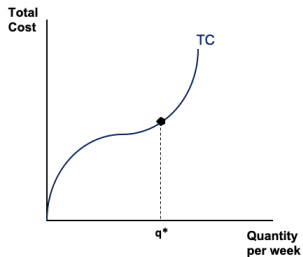
Firms with efficient scale

Shape of AC and MC curves



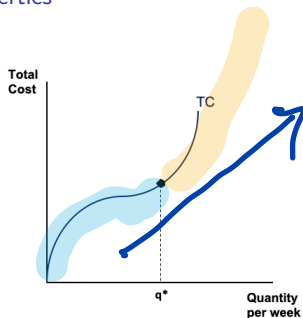
Firms with efficient scale

Properties

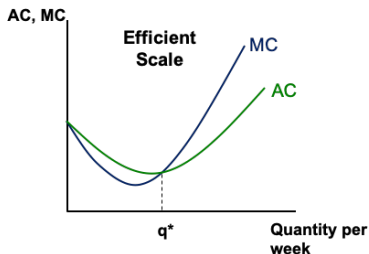


Firms with efficient scale

Properties

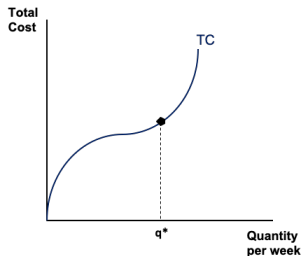


The total costs are always rising, first at a decreasing rate (concave), and then at an increasing rate (convex).

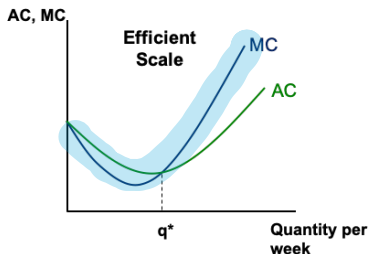


Firms with efficient scale

Properties



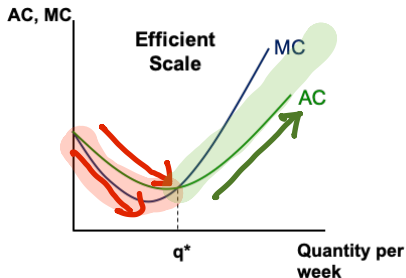
The total costs are always rising, first at a decreasing rate (concave), and then at an increasing rate (convex).



MC (slope of TC) is always positive – it falls till a certain point and then rises.

MC-AC relationship

Firms with efficient scale



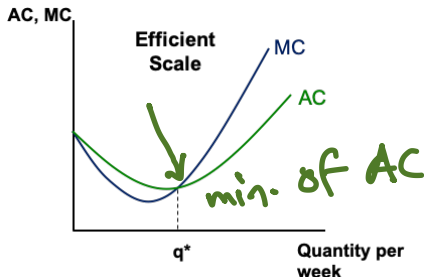
69 80
When $MC < AC$, then $AC \downarrow$

The cost of each additional unit is low, and hence the AC of increasing output is falling.

$MC > AC$, then $AC \uparrow$
93 80

MC-AC relationship

Firms with efficient scale



When $MC < AC$, then $AC \downarrow$

The cost of each additional unit is low, and hence the AC of increasing output is falling.

When $MC > AC$, then $AC \uparrow$

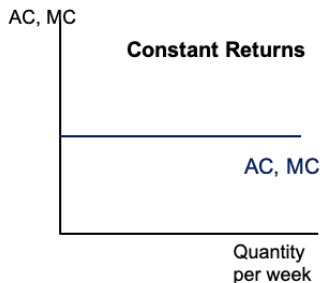
The cost of each additional unit is high, and therefore the AC of increasing output is rising.

Average and Marginal costs

Different scenarios

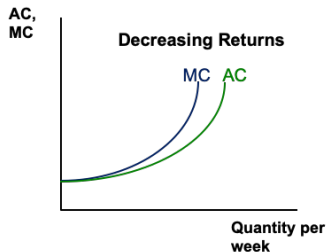
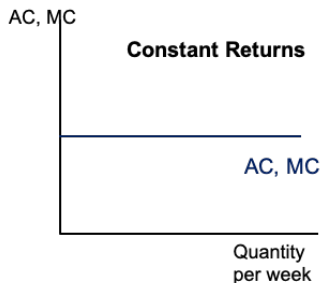
Average and Marginal costs

Different scenarios



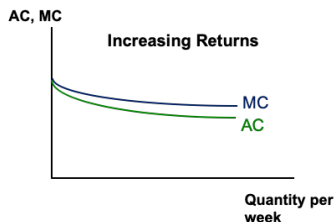
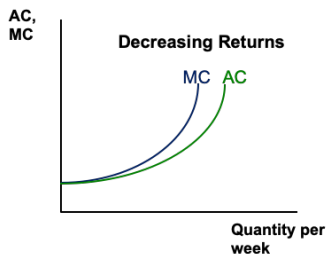
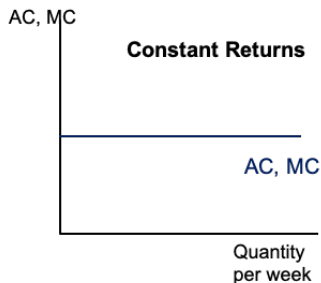
Average and Marginal costs

Different scenarios



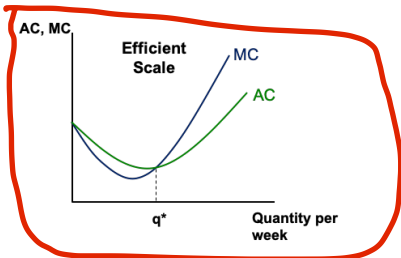
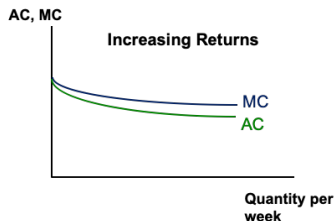
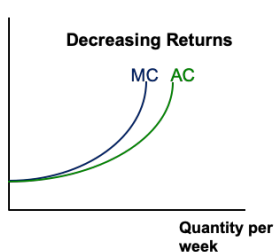
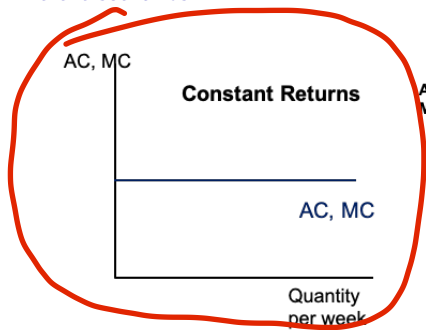
Average and Marginal costs

Different scenarios



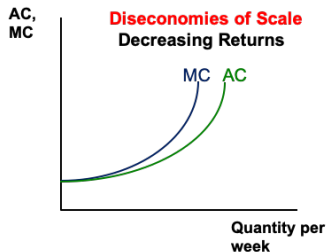
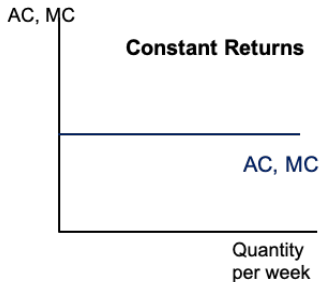
Average and Marginal costs

Different scenarios

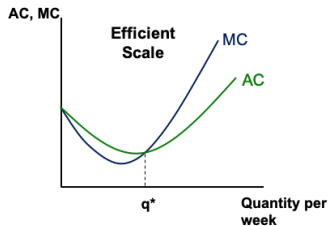
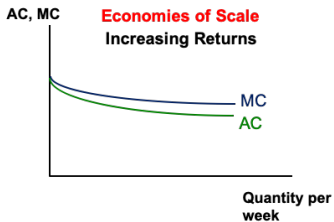
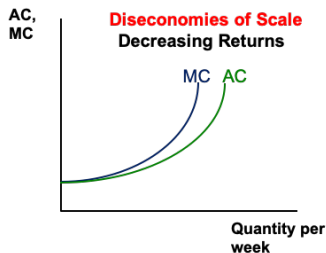
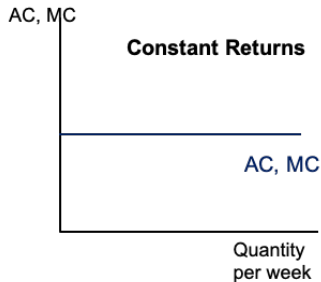


Economies of Scale

Economies of Scale



Economies of Scale



How are *returns to scale* different from *economies of scale*?

Try Yourself!

How are *returns to scale* different from *economies of scale*?

Try Yourself!

Returns to scale refers to how outputs change when all inputs are changed in the same proportion. That is, if output more than doubles when all inputs are doubled, then there are increasing returns to scale.

Economies of scale refers to a property of the cost function – that is whether average costs (AC) rise or fall when more output is produced.

So are these two concepts related?

Relation between *returns to scale* and *economies of scale*

If the production function exhibits increasing returns to scale at all levels of output, then the AC is declining at all points and hence it is a situation with economies of scale.

If the production function exhibits decreasing returns to scale at all levels of output, then the AC is increasing at all points and hence it is a situation with diseconomies of scale.

If the production function exhibits constant returns to scale, then the AC is constant at all points and hence it is neither economies nor diseconomies of scale.